Dr. Vahram Martirosyan

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Faculty Application — Ocean University of China (中国海洋大学), Qingdao
Interpretable Machine Learning for Ocean Science · HPC & Compiler Engineering · Explainable AI

Profile & Research Summary

Ph.D. in Computational Mathematics with **20+ years** spanning academic research, university teaching, and industrial software engineering. My research programme develops **interpretable machine-learning models for ocean and climate science**: stagewise-regularized polynomial regression ensembles that reconstruct global ocean temperature, salinity, and dissolved-oxygen fields and global significant wave height with accuracy competitive with tree ensembles and deep learning — but with explicit physical interpretability, microsecond-to-millisecond inference, and kilobyte-to-sub-megabyte footprints suitable for **on-board, edge, and autonomous deployment at sea**.

Complementary expertise in **high-performance computing and compiler engineering**: LLVM-based automatic parallelization (hpc++) and binary lifting of legacy numerical codes (LEAP), directly applicable to ocean-model acceleration and marine big-data processing. Broader research spans explainable AI for medicine (multi-class ECG), materials science, remote sensing, and precision agriculture, with **25+ publications** (published and accepted IEEE-indexed conference papers, several under review) and a pending high-entropy-alloy design patent. 15+ years of university teaching (programming, systems, architecture, algorithms) with a track record of integrating research into the classroom. Currently **Assistant Professor and Course Team Leader** at Higher Colleges of Technology, Abu Dhabi, and **Chief Scientist** at ExplainableAI.app.

Research Interests

- **Interpretable ML for ocean science:** polynomial zone ensembles for global temperature, salinity, and dissolved-oxygen profile reconstruction; transparent alternatives to deep learning for ocean climatology.
- **Marine climate & wave modelling:** interpretable surrogates for global significant-wave-height climatology; coastal buoy transfer and uncertainty analysis.
- **Marine data science & edge deployment:** resource-constrained, latency-critical models (microsecond-to-millisecond inference, kilobyte-to-sub-megabyte memory) with int8/int16 quantization for at-sea, on-board, and autonomous platforms; spatial-block and profile-stratified validation.
- **HPC for ocean & climate computing:** LLVM-based automatic parallelization (hpc++), binary lifting of legacy numerical codes (LEAP), heterogeneous CPU–GPU execution.
- **Remote sensing data analysis:** multi-temporal optical and SAR feature engineering; oil-spill detection, crop-type mapping, and land/ocean-surface analysis.
- **Explainable AI (PRMs):** polynomial regression models with fully readable coefficients for medicine, materials science, and agriculture.

Education

Ph.D., Computational Mathematics 2006 – 2009
Yerevan State University

Dissertation: "Construction of an Algorithm for the Pseudoinversion of Bidiagonal Matrices and an Investigation of its Parallelization"

M.Sc., Informatics and Applied Mathematics 2004 – 2006
Yerevan State University

B.Sc., Informatics and Applied Mathematics 2000 – 2004
Yerevan State University

Academic & Teaching Experience

Course Team Leader Jan 2025 – Present
Higher Colleges of Technology, Abu Dhabi, UAE

- Managing and guiding a team of lecturers across multiple HCT campuses for the System Architecture & Integration course.
- Developing and maintaining course materials, assessments, and standardised delivery frameworks.

Assistant Professor, Computer Information Science Jan 2024 – Present
Higher Colleges of Technology, Abu Dhabi, UAE

- Courses: Fundamentals of Programming (Java), System Architecture & Integration, IoT & Security, Data Structures & Algorithms, Machine Learning, Data Science, Statistics & Probability.

Senior Lecturer (Contract) Feb 2021 – Dec 2023
Yerevan State University, Faculty of Informatics & Applied Mathematics

- Courses: Operating Systems. Supervised undergraduate research projects.

Faculty Lecturer (Contract) Jan 2023 – Dec 2023
Université française en Arménie (UFAR), Yerevan

- Courses: Computer Architecture I, C Programming.

Visiting Lecturer (Contract) Aug 2022 – Jan 2023
American University of Armenia, Yerevan

- Course: Introduction to Computer Science.

Informatics Teacher (Contract) Sep 2023 – Dec 2023
PhysMath School After A. Shahinyan, Yerevan, Armenia

- Taught Informatics at a specialist physics-mathematics secondary school.

Founder & Lecturer Jan 2019 – Jun 2023
MutEx School, Yerevan

- Founded and ran an independent programming school; designed curricula and delivered public lectures.

Lecturer Sep 2013 – Dec 2019
Armsoft CS Learning Center, Yerevan

- Course: Operating Systems.

Lecturer Feb 2015 – Jun 2015
Armenian Code Academy, Yerevan

- Course: .NET / C# Programming.

Senior Lecturer Feb 2007 – 2014
Russian-Armenian (Slavonic) University, Yerevan

- Courses: Linux/POSIX Programming, Win32 API, Linux Device Driver Development, GUI Software Development (GTK+), Operating Systems, System Programming, Database Management Systems.

Courses Taught

- **Programming:** Fundamentals of Programming, C Programming, C++ Programming, High-performance computing in C++, .NET/C# Programming, Intro to Computer Science.
- **Systems & Architecture:** System Architecture & Integration, Computer Architecture I, Operating Systems, Linux/POSIX Programming, Win32 API Programming, Linux Device Driver Development.
- **Data & Mathematics:** Data Structures & Algorithms, Discrete Mathematics, Applied Discrete Mathematics, Database Management Systems, Statistics & Probability.
- **Machine Learning & Data Science:** Machine Learning, Data Science, Applied Machine Learning, Data Analysis & Visualisation.
- **Emerging Topics:** IoT & Security, Computational Thinking, GUI Development (GTK+), System Programming.

Research & Industry Experience

Research

Chief Scientist Jan 2026 – Present
ExplainableAI.app

- Directing scientific strategy and model architecture for a no-code ML platform built entirely on intrinsically interpretable polynomial regression models — no black-box neural networks.
- Platform combines browser-side WebAssembly training (data never leaves the device), a hand-written C++ inference engine with sub-millisecond latency, and an AI App Builder (powered by Gemini) that turns any trained model into a shareable live web app.
- Applicable to any domain: users train models on any CSV or image dataset and deploy them as shareable live apps — no code, no infrastructure required.
- Authored 15+ research papers in 2026 advancing the theory and application of PRMs across computer vision, healthcare (multi-class ECG), materials science, precision agriculture, ocean/climate monitoring, and structural engineering; several under review, one published and two accepted at IEEE-indexed conferences, plus a pending high-entropy-alloy design patent.
- Delivered production-oriented results: interpretable multi-variable ocean models running in 1.09 ms on an ESP32, a 657 KB Sentinel-1 oil-spill detector at 11,500 tiles/s, and a 123M-parameter ECG classifier compressed 99.7% (1.3 MB) with negligible loss.

Chief Executive Officer Jun 2022 – Jul 2023
System View Armenia, Yerevan

- Developed LLVM/Clang-based auto-vectoriser for Xilinx ACAP (Visual System Integrator toolchain); peak speedup 2009× on GPU, 32.1× on CPU.

Head of System Programming Laboratory Jan 2009 – Jul 2013
Russian-Armenian University / ISP RAS, Yerevan

- Led RAS System Programming Lab; directed Linux Foundation verification of system libraries and kernel modules; managed a team of researchers and engineers.

Junior Research Scientist 2003 – 2005
Institute of Mathematics, National Academy of Sciences of RA

- Parallelisation of numerical algorithms for integral equations using MPI.

Industry (Selected)

Director of Software Engineering Sep – Dec 2023
Plat.AI, Yerevan

Associate Software Development Manager Oct 2021 – Jun 2022
HelpSystems Armenia, Yerevan

Principal Software Engineer / Team Lead Feb 2017 – Oct 2021
HelpSystems Armenia, Yerevan

Software Architect (Contract) 2018 – 2020
Priotix, Yerevan · smart-home IoT system

Principal / Senior Software Developer Aug 2015 – Feb 2017
Optym, Yerevan · logistics optimisation

Senior Software Developer Aug 2013 – Jul 2015
ArdInnoTekh LLC, Yerevan

Software Developer Oct 2006 – Dec 2008
ISMoTech, Yerevan · firmware, C++/C#, ASP.NET

Software Developer Mar 2005 – Oct 2006
Lycos Europe · PHP/MySQL

Publications

Ocean & Climate Data Science

- **Martirosyan, V.** "Efficient and Interpretable On-Device Polynomial Inference for Multi-Variable Ocean Profiling." *eICocean 2027* (IEEE-indexed conference, accepted), 2026.
  ($R^2 = 0.973$ temperature, 0.794 salinity, 0.895 dissolved oxygen; 4.4 µs desktop inference; 1.09 ms on ESP32 and 1.49 ms on RP2350; 180 µJ/96 µJ per inference.)

- **Martirosyan, V.** "Interpretable Polynomial Zone Ensembles for Global Ocean Temperature and Salinity Profile Reconstruction." *Under review, Q1 journal (TGRS / JAMES / Comput. Geosci.)*, 2026.
  (WOA 2018, 534,090 samples; pooled $R^2 = 0.9703$ temperature / 0.8376 salinity; sub-millisecond inference, sub-megabyte footprint.)

- **Martirosyan, V.** "Depth-Zone Polynomial Regression for Global Ocean Climatology." *Under review, IEEE journal (Q1 attempt)*, 2026.
  (IAP T/S Climatology 2024; elastic-net degree-6 models per zone: mixed layer, thermocline, deep ocean; spatially adaptive subsampling 534K → 177K rows.)

- **Martirosyan, V.** "Auditable Kilobyte-Scale Surrogates for the Global Significant-Wave-Height Climatology." *Prepared for IEEE JSTARS*, 2026.

($R^2 = 0.945$, RMSE 0.262 m; cell-holdout $R^2 = 0.933$; coastal R^2 improves from 0.34 to 0.52 with in-situ buoys.)

- **Martirosyan, V.** "Portability and Limits of an Interpretable Polynomial Framework: A Global Significant-Wave-Height Case Study." Working paper, 2026.
(Anomaly $R^2 = 0.875$; offshore buoy RMSE ≈0.5 m; negative result reported for a distance-to-coast feature.)

- **Martirosyan, V.** "Deep-Ocean Temperature Profile Reconstruction via Interpretable Polynomial Zone Ensembles." Working paper, 2026.
(Stagewise-Rank-Polynomial-Regression; 5,005-term basis; 2.6 µs per-profile inference in C++; edge-deployment analysis.)

- **Martirosyan, V.** "Global Ocean Salinity Profile Reconstruction with Interpretable Polynomial Zone Ensembles." Working paper, 2026.
(WOA dataset; degree-8 expansion, 2,001 coefficients per zone; 23 µs inference; river-mouth salinity correction; residual ensemble.)

- **Martirosyan, V.** "Blended Ocean Visualization: Combining Sparse Polynomial Regression with Residual Fields." Working paper, 2026.

- **Martirosyan, V.** "Towards On-Board SAR Oil-Spill Detection: A Proof-of-Concept TinyML Study on Flight-Class Microcontrollers." Submitted to *IEEE eICocean 2027 (Qingdao)*, 2026.
(657 KB → 82 KB int8 model at G-mean 0.822; 15 KB model at G-mean 0.803 fits an ATmega328P; 3.76–9.2 ms/tile on real microcontrollers.)

High-Performance Computing & Compilers

- **Martirosyan, V.** "LEAP: Lifting Legacy x86-64 Binaries to Recompilable IR for Automatic Parallelization." *Under review, ITCA 2026*, 2026.
(43-program liftability study: 100% at `-O0` and 98% at `-O1/O2/O3`; 3.7× geomean speedup from source IR; uplifted binary 10.3× over `-O0` and 2.1× over `-O2`; 246× peak GPU offload; 377/377 tests pass.)

- **Martirosyan, V.** "hpc++: An LLVM-Based Automatic Parallelization System for C++ with GPU Acceleration." *arXiv*, Dec 2025.
(7 parallelisation strategies, zero source modification, 24 benchmarks; 16.8× average, 97× peak CPU, 247× peak GPU; 100% numerical correctness.)

Explainable AI: Medicine, Materials & Remote Sensing

- **Martirosyan, V.** "Stagewise Regularized Polynomial Regression: A Hybrid Optimization Strategy for Medical Diagnosis on Small Datasets." Prepared manuscript, Feb 2026.
(86.7% accuracy on UCI Cleveland Heart Disease; 93% model compression; ≈36 FLOPs/prediction.)

- **Martirosyan, V.** "Polynomial Regression Models for Interpretable Machine Learning: Theory, Optimization, and Medical Applications." Prepared manuscript, Feb 2026.

- **Martirosyan, V.** "Heart Disease Diagnosis Using Third-Order Polynomial Regression Models with L2 Regularization." Prepared manuscript, Feb 2026.
(81.8% accuracy, 10-fold CV; full interpretability via explicit polynomial coefficients.)

- **Martirosyan, V.** "Efficient Multi-Class ECG Classification Using Polynomial Regression with an Analytical Solver." *Applied Computational Intelligence and Soft Computing* (major revision), 2026.
(82.75% ± 1.03% six-class accuracy, macro-F1 0.758; 12 µs inference; outperforms CNN, random forest, XGBoost and SVM baselines.)

- **Martirosyan, V.** "Automated Multi-Class ECG Arrhythmia Classification Using an Ensemble of Polynomial Regression Models." *IEEE ICCA 2026* (accepted), 2026.
  (78.32% balanced accuracy, 68.23% macro-F1; sparsification reduces 123.1M terms to 330K — 1.3 MB, a 99.7% reduction — within ≈1 pp.)

- **Martirosyan, V.** "Explainable and Efficient Digit Classification via Polynomial Regression Networks: A Transparent Alternative to Deep Learning." *arXiv*, Jan 2026.
  (96.85% average MNIST accuracy; 84% parameter reduction at <1% accuracy loss.)

- **Martirosyan, V.** "Accuracy and Interpretability from a Single Solver: Polynomial Regression up to Degree 5 for Concrete Compressive Strength Prediction." *ICACTE 2026* (IEEE-indexed conference), 2026.
  (Degree-5 $R^2 = 0.859$, RMSE 6.25 MPa; 88% term reduction with 74% of the retained terms aligning with known concrete physics; single closed-form solve, no gradient descent.)

- **Martirosyan, V. and N. Martirosyan** "Physics-Informed Polynomial Regression for Certified Concrete Mix Design." *Under review, ICACTE 2027*, 2026.
  (Physics-informed degree-3 SRPR: $R^2 = 0.919$, RMSE 4.73 MPa; certified global inverse design with 10.6–14.5% cost and 16.7% CO_2 savings versus the best measured mixes; interpretable SCM-substitution design rules.)

- **Martirosyan, V.** "Efficient Crop Type Mapping via Degree-2 Polynomial Feature Expansion of Multi-Temporal Optical and SAR Data." Submitted to *IEEE TGRS*, 2026.
  (98.67% test accuracy and macro-F1; 2.3 s training on 16 cores; 17× SIMD speedup.)

- **Martirosyan, V.** "Sparse Polynomial Regression for Interpretable Yield-Strength Prediction in High-Entropy Alloys." *Under review, IEEE ITT 2026*, 2026.
  (Degree-3 $R^2 = 0.738$ on $n = 1{,}014$, statistically tied with random forest and XGBoost and significantly above a neural network (0.618); 8 of 11 retained terms map to known hardening mechanisms; BCC-only $R^2 = 0.841$.)

Systems Verification & Software Engineering

- **Martirosyan, V. et al.** "Testing of Linux File System Drivers." *Proceedings of ISP RAS, vol. 23, Moscow*, Dec 2012.

- **Martirosyan, V. et al.** "The Spruce System: Quality Verification of Linux File System Drivers." *SYRCoSE 2012*, May 2012.

- **Martirosyan, V. et al.** "Dynamic Verification of Linux File System Drivers." *CSIT 2011*, Sep 2011.

- **Martirosyan, V. et al.** "The Spruce System: Linux Kernel Module Verification Automation (in Russian)." *7th Annual Conference of RAU*, Dec 2012.

- **Martirosyan, V. et al.** "Fault Simulation in Linux Kernel Module Verification (in Russian)." *7th Annual Conference of RAU*, Dec 2012.

- **Martirosyan, V. et al.** "Linux Verification – Challenge of Today." *7th Annual Conference of RAU*, Dec 2012.

- **Martirosyan, V. et al.** "X2C – XML To Code: Advantages of Standards." *4th Scientific Annual Conference of RAU*, Dec 2009.

- **Martirosyan, V. et al.** "X2C: Code Generation Technology." .

Books & Tutorials

- **Martirosyan, V.** *Programming via Operating Systems, Second Edition. Hilfmann Press*, Sep 2024.
- **Martirosyan, V. et al.** *Tutorial: Basics of Programming in OS Linux Environment (in Russian). MIPT*, Apr 2011.
- **Martirosyan, V. et al.** *Tutorial: Programming in GNU/Linux Environment – From System Calls to the GUI (in Russian). RAU Publishing*, Dec 2010.

Translations

- *Jinnetic Engineering by Richard Stallman, translated into Armenian.* Dec 2010.

Technical Skills

- **Interpretable ML & Ocean Data:** Polynomial Regression Models (PRMs), stagewise/elastic-net regularization, feature importance; World Ocean Atlas 2018, IAP T/S Climatology 2024, EN4, global wave reanalysis; salinity/dissolved-oxygen/wave-height reconstruction; profile-stratified, spatial-block, and temporal-holdout validation.
- **Edge / On-Device AI:** int8/int16 post-training quantization, fixed-point inference kernels, sub-MB models for microcontrollers and marine edge hardware (ESP32, RP2350, Arduino); latency and energy budgeting.
- **Machine Learning:** Supervised learning, regression modelling, interpretable model design, Python, PyTorch.
- **HPC & Parallelism:** LLVM/Clang (compiler pass development), binary lifting, OpenCL, MPI, POSIX Threads, auto-vectorisation, heterogeneous CPU-GPU execution, Xilinx ACAP.
- **Systems Programming:** C/C++ (expert), Linux kernel modules, device drivers (Ext4, XFS, JFS, BTrFS), WIN32 API, GTK+, IoT/embedded systems.
- **Web & Databases:** PHP/MySQL, ASP.NET/C#/MS SQL Server, SQL.
- **Tooling:** Git, Linux/Unix, LaTeX.
- **Languages:** Armenian (native), Russian (fluent), English (fluent).

Professional Development & Certifications

AI, Machine Learning & Data Science

- Fundamentals of Deep Learning — **NVIDIA**, Apr 2025
- Artificial Intelligence Foundations: Thinking Machines — *LinkedIn*, Jul 2024
- Introduction to Artificial Intelligence — *LinkedIn*, Jul 2024
- Advanced NoSQL for Data Science — *LinkedIn*, Jan 2025
- Statistics Foundations 1: The Basics — *LinkedIn*, Sep 2024

Embedded Systems, FPGA & Hardware

- C Programming for Embedded Applications — *LinkedIn*, Nov 2024
- Introduction to FreeRTOS and Basic Task Management — *LinkedIn*, Nov 2024
- Getting Started with RISC-V — *LinkedIn*, Jul 2024

- Learning Verilog for FPGA Development — *LinkedIn*, Jul 2024
- Learning FPGA Development — *LinkedIn*, Jul 2024
- Introduction to Quantum Computing — *LinkedIn*, Jul 2024

Software Architecture & Engineering

- Software Architecture: Domain-Driven Design — *LinkedIn*, Jul 2024
- Microservices Foundations — *LinkedIn*, Jul 2024
- Microservices: Asynchronous Messaging — *LinkedIn*, Jul 2024
- Microservices: Security — *LinkedIn*, Jul 2024

Leadership & Communication

- How to Be an Effective Manager to Drive Impact — *LinkedIn*, Jul 2024
- Storytelling to Influence Leadership and Decision Makers — *LinkedIn*, May 2024
- Improving Your Thinking — *LinkedIn*, Jul 2024
- Leading with Emotional Intelligence — *LinkedIn*, Apr 2019
- Public Speaking Foundations — *LinkedIn*, Feb 2019